

EngiSense AI — Engineering Test Dataset Documentation

Document ID: ES-TEST-001 | Revision: 1.0 | Synthetic dataset for controlled testing of analytics, RAG, and unified AI.

1. Dataset Context

The CSV represents measurements from a hypothetical rotating engineering test rig. Use the CSV as the engineering dataset and this document as the technical knowledge source for document-aware AI questions.

2. Measurement Definitions

Parameter	Unit	Description
Temperature	°C	Equipment temperature
Voltage	V	DC supply voltage
Current	A	Electrical load current
Vibration	mm/s RMS	Overall vibration level
Pressure	bar	System pressure

3. Operating Limits

Parameter	Normal	Warning	Critical
Temperature	20–42 °C	>42 °C	≥50 °C
Voltage	11.5–12.5 V	<11.5 or >12.5 V	<10.5 or >13.5 V
Current	1.5–4.5 A	>4.5 A	≥5.0 A
Vibration	0–6.0 mm/s RMS	>6.0 mm/s RMS	≥10.0 mm/s RMS
Pressure	0.8–1.5 bar	<0.8 or >1.5 bar	<0.5 or >2.0 bar

4. Interpretation Rules

- A value outside the normal range is outside normal operation.
- A value at or above a critical limit is critical according to this document.
- Statistical outlier detection is separate from engineering operating limits.
- The dataset intentionally contains a vibration spike of 12.5 mm/s RMS at sample 30.
- Voltage is intentionally stable around 12 V.

5. Expected Characteristics

There are 35 observations and 6 columns. Temperature, current, vibration, and pressure generally increase over the sequence. Voltage remains close to 12 V. Sample 30 contains an intentional high vibration reading for outlier testing.

6. Test Questions

- How many observations and columns are present?
- What is the average temperature?
- Which variables are strongly correlated?

- Are there statistical outliers?
- Is sample 30 vibration an engineering-limit violation?
- Is sample 30 vibration a statistical outlier?
- What are the documented vibration limits?
- Does the dataset show a voltage problem?
- What is the difference between a statistical outlier and an engineering-limit violation?

7. Test Note

This is a synthetic software-validation dataset. It must not be used for real operational engineering decisions.